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AUGUST 10, 1956
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FEDERAL SPECIFICATION

WIRE ROPE, STEEL, HIGHWAY GUARD (Highway Guard Cable*)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—This specification covers zinc-coated (galvanized) steel wire ropes for highway guard rails.

1.2 Classification.

1.2.1 *Construction.*—Highway guard wire ropes shall be furnished of the following constructions and diameters, as specified:

3 by 7 (¾ inch) Highway Guard Wire Rope, fig. 1.

7 by 7 (1 inch) Highway Guard Wire Rope, fig. 2.

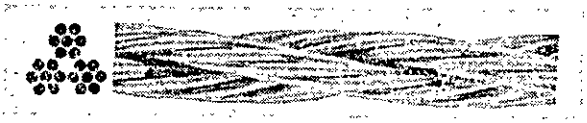


FIGURE 1.—3 by 7 (¾ inch) highway guard wire rope.

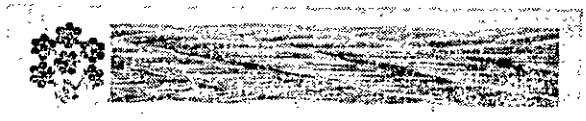


FIGURE 2.—7 by 7 (1 inch) highway guard wire rope.

1.2.2 *Finish.*—Highway guard wire rope shall be of zinc-coated (galvanized) wires.

1.2.3 *Lengths.*—Highway guard wire rope shall be furnished either in the following standard continuous lengths of 500, 1000 and 2000 feet, or cut to the required lengths, as specified.

* Trade name.

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The following specification and standards, of the issues in effect on date of invitation for bids, forms a part of this specification:

Federal Specification:

QQ–M–151—Metal; General Specifications for Inspection of.

Military Standards:

MIL–STD–129—Marking for Shipment and Storage.

MIL–STD–163 — Preparation of Steel Products for Domestic Shipment (Storage and Overseas Shipment).

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(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

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(Copies of Military Standards required by con-

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TABLE I.—*Highway guard wire ropes*

Definitions		Highway guard wire ropes	
		3 by 7	7 by 7
Wires	Number of wires in a wire rope	21	49
	Wire diameter, minimum-maximum, inch	0.117–0.125[1]	0.105–0.113[1]
	Pitch of wires in a strand, maximum, inch	4½	4½
Finished wire rope	Ordered (nominal) diameter, minimum, inch	¾	1
	Maximum diameter, inches	25/32	1 3/64
	Approximate circumference, inches	2⅜	3⅛
	Pitch of strands in a wire rope, maximum, inches	7½	10
	Minimum breaking strength, lb.	25,000	45,000

[1] See 3.3.3.

tractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Wire rope constructions.

3.1.1 *3 by 7 (¾ inch) highway guard wire ropes.*—These wire ropes shall consist of three strands laid around each other (see fig. 1).

3.1.2 *7 by 7 (1 inch) highway guard wire ropes.*—These wire ropes shall consist of six outer strands laid spirally (helically) around a center (core) strand (see fig. 2).

3.2 **Strand construction.**—The strands in the highway guard wire ropes shall be of six outer wires laid spirally (hellically) around a center (core) wire (see figs. 1 and 2).

3.3 Wires.

3.3.1 *Materials.*—Wires in highway guard wire ropes shall be of annealed steel made by open hearth or electric furnace process, and of a chemical composition at the option of the manufacturer, provided that the minimum breaking strengths of the finished wire ropes shall be not less than specified in table I.

3.3.1.1 *Ductility of wires.*—Wires in highway guard wire ropes shall not break nor fracture when closely and tightly wound on a proper mandrel (see 4.4.3).

3.3.2 *Zinc coating (galvanizing).*—The wires shall be uniformly and continuously zinc coated at finished size either by a hot-dip galvanizing or electrodepositing process. In a single wire rope all wires shall be zinc coated by the same process. Unless specified, the selection of the process for zinc coating shall be at the manufacturer's option.

3.3.2.1 *Weight of zinc coating.*—The weight of zinc coating shall be not less than 0.8 oz. sq. ft. when tested by the stripping method (see 4.4.4.1).

3.3.3 *Diameter of wires.*—The zinc-coated wires in the highway guard wire ropes shall be of the diameters within the limits specified in table I. In each wire rope the difference in diameters between individual wires in the strands shall not exceed 0.0045 inch.

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3.3.4 *Joints.*

3.3.4.1 *Joints in wire ropes and strands.*—The highway guard wire ropes and their strands shall be without joints or splices, unless specifically permitted by the purchaser.

3.3.4.2 *Joints in individual wires.*—Wires shall be preferably without joints. When joints are used, the distances between them in a strand shall be not less than 18 inches.

3.3.4.2.1 *Joining methods.* — The joining of individual wires shall be either by lap-brazing or butt-welding. In lap-brazed joints, the length of the lap shall be not less than three diameters of the wires with the overlapping faces completely covered by the brazing metal, smooth, and clean. In butt-welded joints, the wire ends shall not show any signs of overheating, which may affect the material of the wire. All butt-welded joints shall be stress relieved. All wire joints shall be thoroughly coated with zinc or suitable zinc bearing alloy.

3.4 Fabrication.

3.4.1 *Preformed or nonpreformed highway guard wire ropes.*—These wire ropes shall be preformed or nonpreformed, as specified (see 6.3(6)). When the invitation for bids, contract or order does not specify whether the wire ropes shall be preformed or nonpreformed, the contractor may supply either kind.

3.4.2 *Lays.*—Highway guard wire ropes shall be of right regular lay. The wires in the strands shall be laid to the left and each strand to be laid to the right (see 6.3(4)) and figs. 1 and 2).

3.4.3 *Pitches.*—The wires in the strands and the strands in the highway guard wire ropes shall be closely laid with uniform pitches (see 6.3(5)). Strand and wire pitches shall conform to the values stated in table I.

3.5 Dimensional and strength requirements.—Highway guard wire ropes covered by this specification shall conform to the dimensional and breaking strength requirements stated in table I.

3.6 Workmanship.

3.6.1 Highway guard wire ropes shall be securely laid and free from kinks, splits, loose wires, or other defects which may be detrimental to their serviceability and appearance.

3.6.2 Zinc coating on wires shall be continuous, free from uncoated spots, scales, lumps, pits, blisters, gritty areas, dross, flux, and other imperfections non-consistent with good commercial practice.

4. SAMPLING, INSPECTION AND TEST PROCEDURES

4.1 Lot.—Unless otherwise specified, a lot shall consist of all highway guard wire ropes of the same construction, produced under approximately the same conditions, by the same manufacturer and presented for delivery at one time on the same contract or order.

4.2 Sampling.—For sampling purposes, all wire ropes in the lot shall be segregated, if possible, by their manufacturing reels.

4.2.1 *When highway guard wire ropes on shipping reels can be identified by their manufacturing reel.*—The wire rope on one shipping reel, taken at random from all reels with the wire ropes cut from the same manufacturing reel, shall be subject to inspection and tests.

4.2.2 *When highway guard wire ropes on shipping reels cannot be identified by their manufacturing reel.* — Wire ropes on each shipping reel shall be subject to inspection and tests.

4.3 Inspection.

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4.3.1 *Determination of the highway guard wire rope diameter.*—The wire rope diameter see 6.3(2), shall be determined at the free end of the wire rope on the shipping reel under inspection. Beginning not less than 5 feet from the end of the wire rope, the diameter shall be measured at three places not less than 5 feet apart with a slide caliper or with any other proper instrument. The average of the measured diameters shall be the diameter of the wire rope under inspection.

4.3.2 *Determination of the wire and strand pitches.*—The wire and strand pitches, see 6.3(5), shall be determined at the free end of the wire rope on the shipping reel under inspection. Beginning not less than 10 feet from the end of the wire rope, the length of five or more of the pitches shall be measured. The distance measured shall be divided by the number of pitches in order to arrive to the pitch size.

4.3.3 *Determination of the wire diameter.*—The diameters of individual wires shall be measured for compliance with the requirements stated in 3.3 and table I.

4.4 **Tests.**

4.4.1 *Test samples.*—One test sample 12 feet long shall be taken for tests from each sample shipping reel as specified in 4.2.1 and 4.2.2.

4.4.1.1 These 12-foot length samples shall be in addition to the lengths of the wire ropes specified in the invitation for bids, contract or order.

4.4.1.2 The 12-foot sample length shall be seized and cut into two lengths—a 9-foot length and a 3-foot length.

4.4.2 *Breaking strength test.* — Breaking strength of the wire rope shall be determined from the long samples. The free length of the wire rope under the test shall be not less than 2 feet. The head of the testing machine shall be moved at a rate of not more than 3 inches per minute. Should a specimen not meet the breaking strength requirements and the break occurs within 1 inch of the grips, another test sample may be prepared and tested.

4.4.3 *Test for ductility of wires (wrapping test).*—From each short sample of wire rope, not less than one wire specimen from each strand shall be taken. The wire specimens shall remain as unwound from the sample. One end of the wire specimen shall be secured on the test mandrel in any convenient way. The diameter of the test mandrel shall be three times the diameter of the wire. The test specimens shall not break nor fracture when they are closely and tightly wound on the mandrels at a rate of not more than 15 turns per minute for not less than 6 turns.

4.4.4 *Test for the weight of the zinc coating.*

4.4.4.1 *Specimens.*—A minimum of three wire specimens about 24 inches in length, shall be randomly taken from each short test sample.

4.4.4.2 *Cleaning.*—The specimens shall be cleaned with a suitable volatile organic solvent, then with alcohol and dried thoroughly with a clean cotton cloth. After cleaning, the wire specimen may be loosely coiled to fit the stripping bath.

4.4.4.3 *Stripping solution.*—Twenty grams of antimony trioxide (Sb_2O_3), or 32 grams of antimony trichloride ($SbCl_3$) shall be dissolved in one liter of concentrated hydrochloric acid (HCl) (sp. gr. = 1.19). Five milliliters of one of these antimony chloride solutions shall be added to each 100 milliliters of concentrated hydrochloric acid (sp. gr. = 1.19) to produce the stripping solution.

4.4.4.4 *Procedure.*—The cleaned specimens shall be weighed to the nearest 0.01 gram, and then completely immersed in the stripping solution and held therein, until the violent chemical action ceases and only a few bubbles are being evolved. The temperature of the solution shall at no time exceed 100°F. (38°C). Not more than three wire specimens

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shall be immersed per 100 milliliters of the stripping solution. As soon as the violent chemical action ceases the specimens shall be removed, washed thoroughly, wiped dry and weighed to the nearest 0.01 gram. The diameter of the stripped specimens shall be determined to the nearest 0.001 inch by taking the average of two measurements at right angle to each other. The same stripping solution may be repeatedly used without further additions of antimony chloride solution, until the time required for stripping becomes inconveniently long.

4.4.4.5 *Calculations.* — The weight of the stripped zinc coating shall be determined as follows:

$$A = \frac{W_1 - W_2}{W_2} \times d \times 163$$

Where: A = weight of zinc-coating in ounces per square foot of stripped wire surface.

W_1 = original weight of specimen, in grams.

W_2 = weight of stripped specimen, in grams.

d = diameter of stripped wire in inches.

163 = constant.

4.4.5 *Fabrication test.*

4.4.5.1 *Determination of preforming.* — Short samples shall be used and the seizing wires on both ends should be removed. In preformed wire ropes the strands and wire ends will remain as seized, whereas, in non-preformed wire ropes they will spring apart. The strands shall be now unraveled. In the preformed wire ropes, the unraveled strands will retain the spiral form they had in the wire ropes, and can be put back together again. In non-preformed wire ropes, the unraveled strands change their spiral form they had in the wire rope, and cannot be reassembled. The same test is applicable to the unraveled strand wires.

4.4.6 *Rejection and retesting.*

4.4.6.1 *Rejection.*—If a wire rope (or a specimen cut from it), representing one manufacturing reel, fails to conform to the requirements of this specification, then all shipping reels with the wire rope cut from this manufacturing reel shall be rejected. If a wire rope (or a specimen cut from it), represents an individual shipping reel and fails to conform to the requirements of this specification, then the particular shipping reel shall be rejected.

4.4.6.2 *Retesting.*—The rejected wire ropes at the request of the manufacturer may be reinspected and retested in accordance with the regulations in Federal Specification QQ–M–151. One reinspection and retest only shall be permitted.

5. PREPARATION FOR DELIVERY

(For definitions and levels see par. 6.4).

5.1 The purchasing agencies shall define in their invitations for bids, contract or orders the levels of preservation, packaging and packing of the material.

5.1.1 *Level A (packing for domestic (storage) and overseas shipment).*—Unless otherwise specified, highway guard wire ropes shall be packed and shipped in accordance with the regulations in Military Standard MIL–STD–163.

5.1.2 *Level C (packing for domestic shipment, immediate use).*—Highway guard wire ropes shall be shipped on wooden reels and in continuous lengths, as specified. The shipments shall be prepared in accordance with good commercial practice and in such a manner as to insure acceptance and safe delivery to the designated point by common and other carriers at the lowest rate of the carrier.

5.2 **Marking.**—Each reel shall be legibly and permanently marked or carry an oil proof or metal tag with the following information:

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Federal Specification Symbol: RR–W–420
HIGHWAY GUARD WIRE ROPE
Construction and diameter
Length
Name of the manufacturer
Name of the contractor (if different from the manufacturer's name)
Number of the order or contract
Name of the consignee
Gross weight, in pounds.

5.2.1 When specified, the shipping reels on Army, Navy and Air Force orders, shall be marked or tagged in accordance with the Military Standard MIL–STD–129.

6. NOTES

6.1 Intended use.—7 by 7 (1 inch) highway guard wire ropes are for use mostly at the top of highway guard railings. Whereas, the 3 by 7 (¾ inch) highway guard wire ropes are for general use in highway guard railings.

6.2 Ordering data.—Purchasing agencies should specify: Federal Specification symbol, title and date; construction and diameter of the highway guard wire rope; continuous length (in feet) of the required wire rope; number of lengths required; level of packing; and should exercise any desired option offered herein (see 1.2.3, 3.3.2, 3.4.1, 4.1, 5.1.1 and 5.2.1).

6.3 Definition of terms used.

(1) *Breaking strength.* — The measured load required to pull apart the wire rope in tension.

(2) *Wire Rope diameter (Ordered or Nominal).*—The measured diameter of the circle circumscribing all strands in the wire rope.

(3) *Circumference.* — The measured perimeter of the circle circumscribing all strands in the wire rope.

(4) *Lay.*—The manner in which the outer wires in a strand and the outer strands in a wire rope are laid (twisted) around a center (core) wire or strand respectively.

(5) *Pitch.* — The distance, parallel to the axis of the strand or wire rope, in which a wire or a strand makes a complete turn. (The pitch is also called—"the lay length").

(6) *Preformed or nonpreformed wire ropes.*—Preformed wire ropes are wire ropes in which the wires in the strands and the strands in the ropes were, prior to the last closing operation, permanently shaped to the spiral form (helix) they assume in the strands and wire ropes. Nonpreformed wire ropes are wire ropes which are fabricated from straight wires and strands. Preformed wire ropes are more flexible and less apt to unlay when cut from the reel.

(7) *Reels, Manufacturing.*—Manufacturing reels are the reels on which the wire ropes are wound as they are fabricated on the rope closing machines.

(8) *Reels, shipping.* — Shipping reels are wooden reels on which the wire ropes, cut to the ordered lengths, are wound from the manufacturing reel for shipment.

(9) *Seize (or Serve).*—To bind securely the ends of the wire ropes with a soft annealed iron wire (seizing wire).

6.4 Federal Standard No. 102 should be referred to for definitions and applications of the various levels of packaging protection for supplies and equipment.

6.5 Transportation description.—The applicable transportation description for highway guard wire ropes is as follows:

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Wire Rope, Iron or Steel.

Carload weight, minimum 40,000 pounds.
Truckload weight, minimum 36,000 pounds.

6.6 This specification is based in the main on the following specifications and test standards promulgated by the American Society for Testing Materials (ASTM) and by the American Association of State Highway Officials (AASHO):

AASHO: M30–49.—Standard Specification for Wire Rope and Fittings for Highway Guard Rail.

AASHO: T39–42.—Standard Method of Test for Wire Rope (Guardrail).

ASTM: A90–39 } Standard Methods of
AASHO: T65–55 } Test for Weight of Coating on Zinc-coated (galvanized) Iron or Steel Articles.

6.7 It is believed that this specification adequately describes the characteristics necessary to secure the desired highway guard wire ropes and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids, and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.8 Federal specifications do not include all types, classes, sizes etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, classes, etc., which are suitable for Federal Government requirements.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility, nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—T E
Air Force.

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